

Day 3.2 – Enterprise DNS, DHCP, SRV Records, and DDNS Troubleshooting

Date: May 30–31, 2026

Objectives

- Configure Enterprise DNS using BIND9
- Configure DHCP using OPNsense Kea DHCP
- Implement forward and reverse DNS
- Configure LDAP SRV records
- Begin Dynamic DNS (DDNS) integration

DNS Configuration

Forward Zone

Created zone:

```
int.worldskills.ph
```

Configured records:

```
ns1      IN A    10.1.10.10
int-srv01 IN A    10.1.10.10
```

Reverse Zone

Configured PTR record:

```
10 IN PTR int-srv01.int.worldskills.ph.
```

Validation

Forward lookup:

```
dig @10.1.10.10 ns1.int.worldskills.ph
```

Reverse lookup:

```
dig @10.1.10.10 -x 10.1.10.10
```

Results were successful.

DHCP Configuration

Configured OPNsense Kea DHCPv4:

```
Subnet: 10.1.10.0/24
Pool:   10.1.10.100 - 10.1.10.200
Gateway: 10.1.10.1
DNS: 10.1.10.10
Domain: int.worldskills.ph
```

Ubuntu client successfully obtained:

```
IP Address: 10.1.10.100
DNS Server: 10.1.10.10
Search Domain: int.worldskills.ph
```

LDAP SRV Record

Created LDAP service discovery record:

```
_auth._tcp IN SRV 10 50 389 int-srv01.int.worldskills.ph.
```

Verified using:

```
dig @10.1.10.10 SRV _auth._tcp.int.worldskills.ph
```

DDNS Implementation

Completed

Created TSIG key:

```
key "ddns-key" {  
    algorithm hmac-sha256;  
    secret "****REDACTED****";  
};
```

Configured BIND dynamic updates:

```
allow-update { key "ddns-key"; };
```

Configured OPNsense:

```
DNS Forward Zone: int.worldskills.ph  
DNS Server: 10.1.10.10  
TSIG Key: ddns-key  
Algorithm: hmac-sha256
```

Fixed DDNS Agent startup issue by changing:

```
Bind Address: 127.0.0.1  
Port: 53001
```

Current Status

Completed

- Forward DNS
- Reverse DNS
- DHCPv4
- Search Domains
- LDAP SRV Records
- TSIG Key Integration
- DDNS Agent Startup

In Progress

- Automatic DNS registration of DHCP clients via DDNS

Current issue:

```
host ubuntu-server-01.int.worldskills.ph 10.1.10.10
```

returns:

```
NXDOMAIN
```

indicating DDNS updates are not yet reaching BIND.

Progress

Day 3.2 Completion: ~95%

Only the final DDNS client registration workflow remains.